

RESEARCH FOCUS *Machine learning for reasoning and decision-making under uncertainty.*

EDUCATION

2026 **Cornell Tech | New York, NY**  
Doctor of Philosophy in Computer Science

2024 MS in Computer Science, conferred on PhD candidacy | GPA 4.0  
Areas: AI / ML, Scientific Computing, Applied Probability & Statistics  
*NSF Graduate Research Fellow | Presidential Life Science Fellow*

2021 **University of Pennsylvania | Philadelphia, PA**  
Master of Computer & Information Technology | GPA 3.97  
*Interdisciplinary Innovation Fellow | Reproducible Research Fellow*

2016 **Smith College | Northampton, MA**  
BA Anthropology (Biological, Medical), Environmental Science | GPA 3.97  
*Summa Cum Laude – Top 1% of class | Phi Beta Kappa | Sigma Xi*

EXPERIENCE

03.2026 – **Visitor**  
06.2026 *Isaac Newton Institute for Mathematical Sciences | Cambridge, UK*  
Invited **long-residency program** at the University of Cambridge on the theory and methods of causal inference.

05.2025 – **Research Intern**  
08.2025 *YRIKKA | New York, NY*  
*PI: Dr. Kia Khezeli.* Test-time adaptation and world modeling for abstract, causal, and logical reasoning in large language models.  
*Outcomes: NeurIPS LAW 2025 (spotlight), Amazon Trusted AI (poster).*

05.2024 – **Research Intern**  
08.2024 *Microsoft Research (MSR), Machine Intelligence Group | Cambridge, UK*  
*PI: Dr. Aditya Nori, Dr. Javier González.* Methods for evaluating and improving LLM performance on causal and compositional reasoning tasks.  
*Outcomes: ICML 2025, ICLR 2025, pending patent, 2.5k+ HF downloads.*

05.2022 – **Clinical Data Science Intern**  
08.2022 *Boehringer Ingelheim, Biostatistics & Data Sciences | Ridgefield, CT*  
*PI: Dr. Yi Liu.* Internal research on multimodal deep learning for survival analysis in pharmaceutical development.

08.2021 – **PhD Student Researcher**  
Present *Weill Cornell Medicine, Institute of AI for Digital Health | New York, NY*  
*PI: Dr. Fei Wang.* ML and causal inference for computational biomedicine.

*Cornell Tech Computer Science | New York, NY*  
*PI: Dr. Volodymyr Kuleshov.* Deep generative and probabilistic modeling.

*Cornell Tech Operations Research | New York, NY*  
*PI: Dr. Kyra Gan.* Robust and efficient statistical inference, scalable causal discovery, and causal fairness in healthcare.

05.2020 – **Master's Student Researcher**  
07.2021 *University of Pennsylvania Bioengineering | Philadelphia, PA*  
*PI: Dr. César de la Fuente.* New paradigms for ML-based drug discovery.  
*Outcomes:* Thesis coined the new field of *molecular de-extinction*, published in *Cell Host & Microbe* and covered by *NPR*, *Nature News*, and *CNN*.

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| LANGUAGES                                                   | <i>Proficient:</i> Python; R; L <sup>A</sup> T <sub>E</sub> X; shell. <i>Prior experience:</i> Stan; Java; C; MATLAB.                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                            |
| FRAMEWORKS                                                  | PyTorch; NumPy; sklearn; tidyverse; Git; AWS; Slurm-based HPC; <b>ARC-AGI</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                            |
| SKILLS &<br>INTERESTS                                       | <i>2019 – Present:</i> <b>Probabilistic graphical models</b> ; generative models; <b>AI reasoning</b> ; <b>reinforcement learning</b> ; <b>world models</b> ; <b>causal inference</b> ; <b>causal discovery</b> ; <b>causal fairness</b> ; <b>AI evaluation</b> ; <b>test-time adaptation</b> ; graph theory; applied probability; statistics; logic; AI safety; <b>trustworthiness</b> ; <b>computational biomedicine</b> ; <b>drug discovery</b> . <i>Pre-2019:</i> <b>Molecular genetics</b> , <b>molecular diagnostics</b> , <b>epidemiology</b> . |                                                                                                                                                                                                                                                                                                                                                                            |
| UNDER REVIEW                                                | 2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Xu, J; <b>Maasch, J</b> ; Gan, K. <i>Integrating Causal DAGs in Deep RL: Activating Minimal Markovian States with Multi-Order Exposure</i> . [ARXIV]                                                                                                                                                                                                                       |
|                                                             | 2025                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Maasch, J</b> ; Neiswanger, W; Kuleshov, V; Ermon, S. <i>Probabilistic Graphical Models: A Concise Tutorial</i> . [ARXIV] [WEBSITE]                                                                                                                                                                                                                                     |
| SELECT<br>PEER-REVIEWED<br>PUBLICATIONS<br>(GOOGLE SCHOLAR) | 2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | [ICML] Lawrence, R*; <b>Maasch, J</b> *. <i>Position: AI Reasoning Requires Process Validity</i> . *Equal contribution. 43 <sup>rd</sup> International Conference on Machine Learning. [WEBSITE] [SLIDES] [POSTER] [CODE]                                                                                                                                                  |
|                                                             | 2025                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | [ICML] <b>Maasch, J</b> ; Hüyük, A; Xu, X; Nori A; González J. <i>Compositional Causal Reasoning Evaluation in Language Models</i> . 42 <sup>nd</sup> International Conference on Machine Learning. [WEBSITE] [ARXIV] [SLIDES] [POSTER]                                                                                                                                    |
|                                                             | 2025                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | [ICLR - ORAL - TOP 1.8%] Hüyük, A; Xu, X; <b>Maasch, J</b> ; et al. <i>Reasoning Elicitation in Language Models via Counterfactual Feedback</i> . 13 <sup>th</sup> International Conference on Learning Representations. [ARXIV]                                                                                                                                           |
|                                                             | 2025                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | [AAAI] <b>Maasch, J</b> ; et al. <i>Local Causal Discovery for Structural Evidence of Direct Discrimination</i> . 39 <sup>th</sup> Annual AAAI Conference on Artificial Intelligence. [ARXIV] [SLIDES] [POSTER]                                                                                                                                                            |
|                                                             | 2024                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | [NEURIPS] Hiremath, S; <b>Maasch, J</b> ; et al. <i>Hybrid Top-Down Global Causal Discovery with Local Search for Linear and Nonlinear Additive Noise Models</i> . 38 <sup>th</sup> Annual Conference on Neural Information Processing Systems. [ARXIV]                                                                                                                    |
|                                                             | 2024                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | [UAI] <b>Maasch, J</b> ; et al. <i>Local Discovery by Partitioning: Polynomial-Time Causal Discovery Around Exposure-Outcome Pairs</i> . 40 <sup>th</sup> Conference on Uncertainty in Artificial Intelligence. [ARXIV] [SLIDES] [POSTER]                                                                                                                                  |
|                                                             | 2023                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | [CELL HOST & MICROBE] <b>Maasch, J</b> *; Torres, M*; et al. <i>Molecular de-extinction of ancient antimicrobial peptides enabled by machine learning</i> . Cell Host & Microbe. 31. 8. 1260-1274. e6. 2023. *Equal contribution. [CELL]                                                                                                                                   |
|                                                             | 2023                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | [ICML] <b>Maasch, J</b> ; et al. <i>Regularized Data Programming with Automated Bayesian Prior Selection</i> . ICML Workshop on Structured Probabilistic Inference & Generative Modeling. [WORKSHOP] [ARXIV]                                                                                                                                                               |
| PEER-REVIEWED<br>WORKSHOP<br>PRESENTATIONS                  | 2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | [ICLR] Lawrence, R*; <b>Maasch, J</b> *. <i>Position: Beyond Reasoning Zombies — AI Reasoning Requires Process Validity</i> . *Equal contribution. ICLR Workshop on Logical Reasoning of Large Language Models and ICLR Workshop on Principled Design for Trustworthy AI – Interpretability, Robustness, and Safety across Modalities. [WORKSHOP 1] [WORKSHOP 2] [WEBSITE] |
|                                                             | 2025                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | [NEURIPS - SPOTLIGHT] <b>Maasch, J</b> ; Kalantari, J; Khezeli, K. <i>CausalARC: Abstract Reasoning with Causal World Models</i> . NeurIPS LAW: Bridging Language, Agent, and World Models. [WORKSHOP] [ARXIV] [WEBSITE]                                                                                                                                                   |
|                                                             | 2023                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | [NEURIPS] <b>Maasch, J</b> ; et al. <i>Local Discovery by Partitioning: Polynomial-Time Causal Discovery Around Exposure-Outcome Pairs</i> . NeurIPS Causal Representation Learning Workshop. [WORKSHOP] [ARXIV]                                                                                                                                                           |
|                                                             | 2023                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | [ICML] <b>Maasch, J</b> ; et al. <i>Regularized Data Programming with Automated Bayesian Prior Selection</i> . ICML Workshop on Structured Probabilistic Inference & Generative Modeling. [WORKSHOP] [ARXIV]                                                                                                                                                               |
| PENDING PATENTS                                             | 2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | González J; Nori A; Hüyük, A; Xu, X; <b>Maasch, J</b> . <i>Fine-Tuning Language Models for Reasoning with Counterfactual Feedback</i> . App no: 63/699,777.                                                                                                                                                                                                                |

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| SELECT<br>FELLOWSHIPS &<br>AWARDS | 2025                                                                                                                                                                                                                                                                                                             | Doctoral Fellowship   Cornell Tech <b>Digital Life Initiative</b>                                                                                   |
|                                   | 2023                                                                                                                                                                                                                                                                                                             | Outstanding Service and Community Award   Cornell Tech                                                                                              |
|                                   | 2021                                                                                                                                                                                                                                                                                                             | NSF Graduate Research Fellowship   US National Science Foundation                                                                                   |
|                                   | 2021                                                                                                                                                                                                                                                                                                             | Presidential Life Science Fellowship   Cornell University                                                                                           |
|                                   | 2021                                                                                                                                                                                                                                                                                                             | Reproducible Research Fellowship   OKFN, Alfred P. Sloan Foundation                                                                                 |
|                                   | 2020                                                                                                                                                                                                                                                                                                             | Interdisciplinary Innovation Fellowship   University of Pennsylvania                                                                                |
| INVITED<br>PRESENTATIONS          | 02.26                                                                                                                                                                                                                                                                                                            | <b>TALK</b> <i>What is reasoning?</i> [ <b>SLIDES</b> ]<br>Cornell Tech <b>Digital Life Initiative</b>   New York, NY                               |
|                                   | 01.26                                                                                                                                                                                                                                                                                                            | <b>POSTER</b> <i>Stress-Testing Reasoning with Causal World Models.</i> [ <b>WEBSITE</b> ]<br><b>Amazon AGI Trusted AI Symposium</b>   New York, NY |
|                                   | 07.25                                                                                                                                                                                                                                                                                                            | <b>TALK</b> <i>Compositional Causal Reasoning Evaluation.</i><br>Microsoft Expo Booth, ICML   Vancouver, BC                                         |
|                                   | 04.25                                                                                                                                                                                                                                                                                                            | <b>TALK</b> <i>Graphical Perspectives on Causal Reasoning.</i> [ <b>SLIDES</b> ]<br><b>Flatiron Institute</b> , Simons Foundation   New York, NY    |
|                                   | 03.25                                                                                                                                                                                                                                                                                                            | <b>TALK</b> <i>A brief introduction to causal inference.</i> [ <b>SLIDES</b> ]<br>Cornell INFO5375: Machine Learning for Health   New York, NY      |
|                                   | 10.24                                                                                                                                                                                                                                                                                                            | <b>TALK</b> <i>Local discovery for causal fairness analysis.</i> [ <b>SLIDES</b> ]<br>INFORMS Annual Meeting   Seattle, WA                          |
|                                   | 07.24                                                                                                                                                                                                                                                                                                            | <b>TALK</b> <i>Compositional Causal Reasoning.</i><br>Microsoft Research Machine Intelligence Core   Cambridge, UK                                  |
|                                   | 06.24                                                                                                                                                                                                                                                                                                            | <b>TALK</b> <i>Local causal discovery for causal effect estimation.</i><br>University of Cambridge Statistical Laboratory   Cambridge, UK           |
|                                   | 04.24                                                                                                                                                                                                                                                                                                            | <b>TALK</b> <i>Local Discovery by Partitioning.</i> [ <b>SLIDES</b> ]<br>34th Annual POMS Conference   Minneapolis, MN                              |
|                                   |                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                     |
| PROFESSIONAL<br>ACTIVITIES        | 24-25                                                                                                                                                                                                                                                                                                            | Co-organizer, <b>NYC Learning on Graphs Workshop</b>                                                                                                |
|                                   | 24-25                                                                                                                                                                                                                                                                                                            | PhD Application Reviewer, Cornell Computer Science Graduate Admissions                                                                              |
|                                   | 23-25                                                                                                                                                                                                                                                                                                            | Student leader, Cornell CS PhD Visit Days                                                                                                           |
|                                   | 2023                                                                                                                                                                                                                                                                                                             | Co-developer, <b>Cornell CS 6006: Succeeding in the Graduate Environment</b>                                                                        |
|                                   | 2023                                                                                                                                                                                                                                                                                                             | Founder / organizer, <b>Cornell Causal Reading Group</b>                                                                                            |
| TRAVEL &<br>CONFERENCE<br>GRANTS  | 2026                                                                                                                                                                                                                                                                                                             | Research Travel Grant   Cornell Graduate School                                                                                                     |
|                                   | 2025                                                                                                                                                                                                                                                                                                             | ICML, Vancouver, BC   International Conference on Machine Learning                                                                                  |
|                                   | 2025                                                                                                                                                                                                                                                                                                             | AAAI, Philadelphia, PA   Association for the Advancement of AI                                                                                      |
|                                   | 2023                                                                                                                                                                                                                                                                                                             | NeurIPS, New Orleans, LA   Neural Information Processing Systems                                                                                    |
|                                   | 2020                                                                                                                                                                                                                                                                                                             | Grace Hopper Celebration Scholarship   University of Pennsylvania                                                                                   |
| PEER REVIEW                       | AI/ML: ICML (main track, position track, <b>SPIGM</b> ); UAI (main track); AISTATS (main track); ACL (main track); NeurIPS ( <b>WiML</b> ). <i>Biology</i> : Communications Biology (Nature Portfolio); Journal of Biomedical Informatics (Elsevier); Bioinformatics (Oxford Academic); ACS Infectious Diseases. |                                                                                                                                                     |